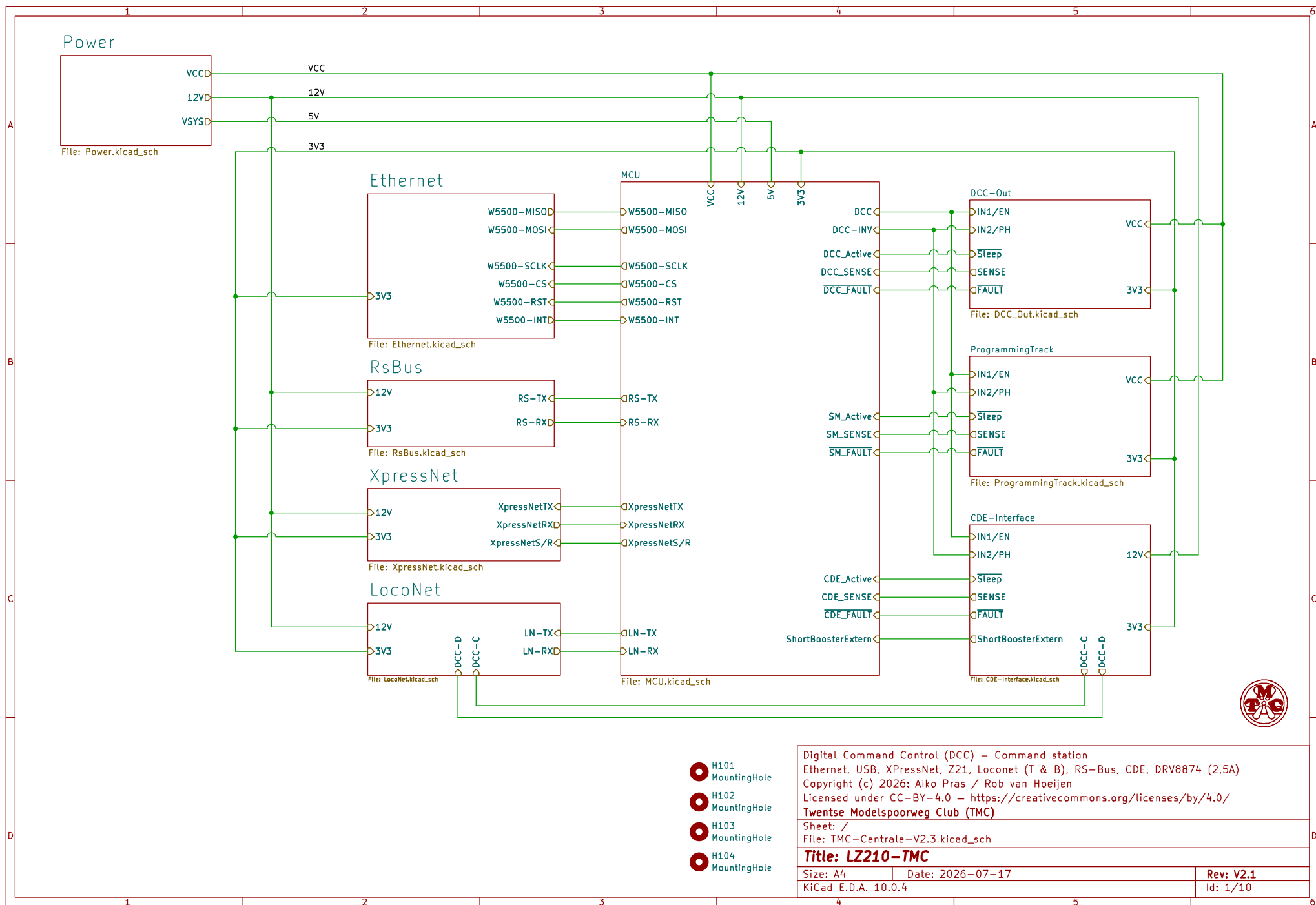
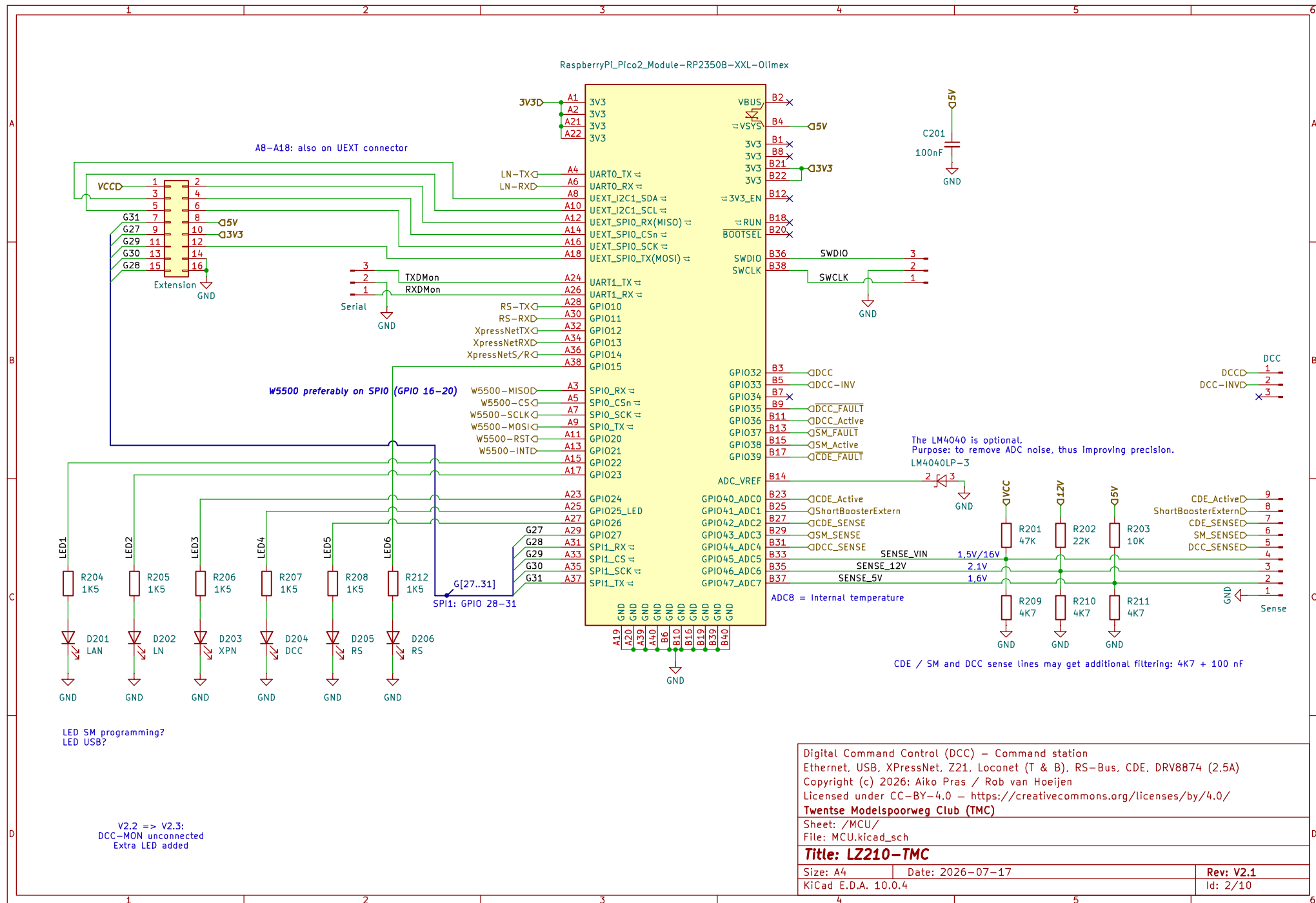
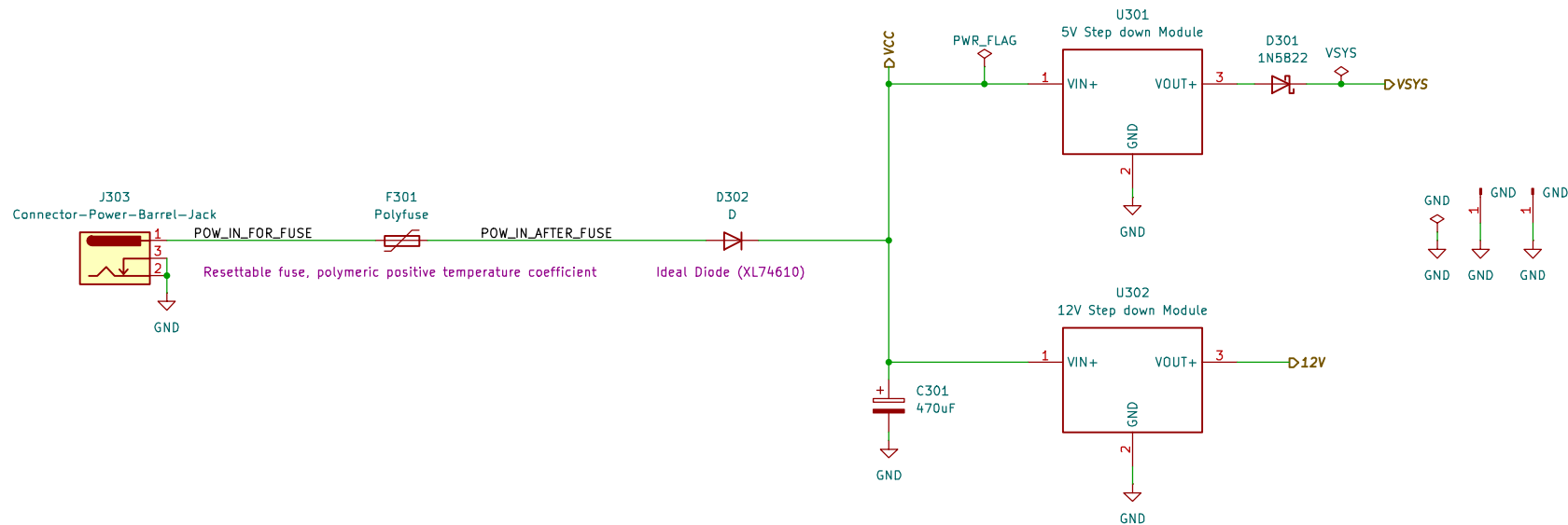








Digital Command Control (DCC) – Command station
 Ethernet, USB, XPressNet, Z21, Loconet (T & B), RS-Bus, CDE, DRV8874 (2,5A)
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Twentse Modelspoorweg Club (TMC)

Sheet: /Power/
 File: Power.kicad_sch

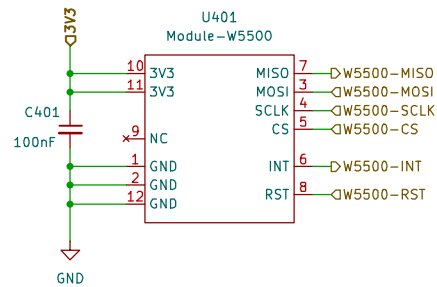
Title: LZ210-TMC

Size: A4 Date: 2026-07-17

Rev: V2.1

KiCad E.D.A. 10.0.4

Id: 3/10



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Sheet: /Ethernet/
File: Ethernet.kicad_sch

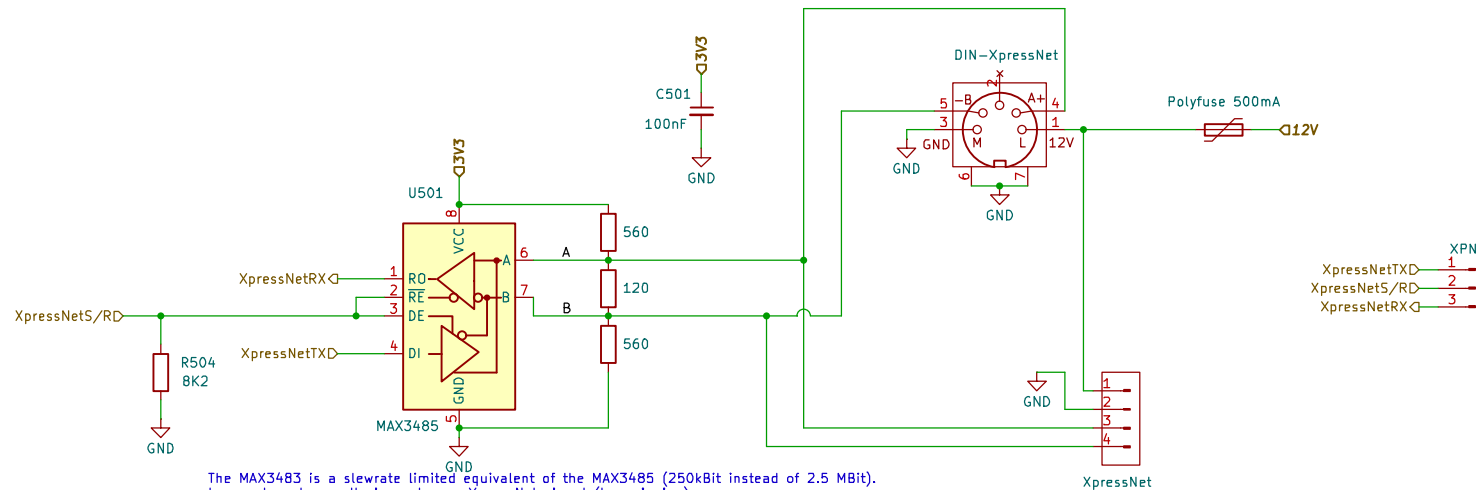
Title: LZ210-TMC

Size: A4 Date: 2026-07-17

Rev: V2.1

KiCad E.D.A. 10.0.4

Id: 4/10



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Sheet: /XpressNet/
File: XpressNet.kicad_sch

Title: LZ210-TMC

Size: A4 Date: 2026-07-17

Rev: V2.1

KiCad E.D.A. 10.0.4

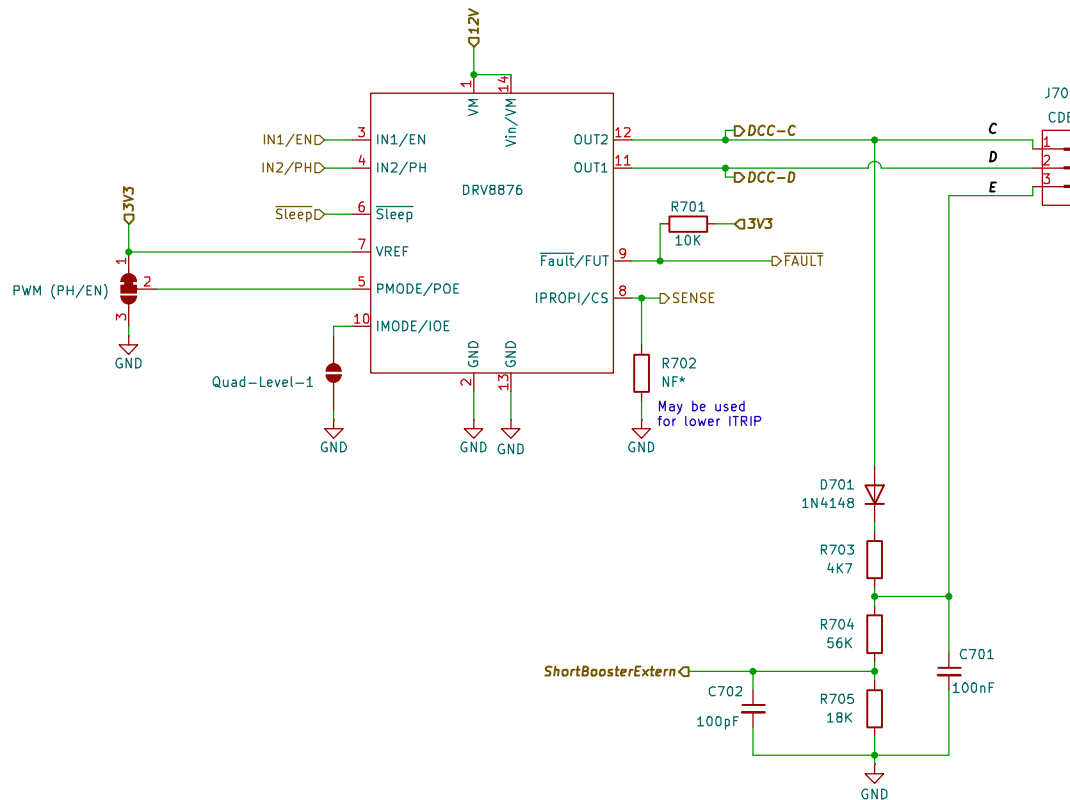
Id: 5/10

PMODE STATE	CONTROL MODE
PMODE = Logic Low	PH/EN
PMODE = Logic High	PWM
PMODE = HI-Z	Independent Halfbridge

*DRV8871: only PWM

IMODE STATE		IMODE FUNCTION		nFAULT Response
		Current Chopping Mode	Overcurrent Response	
Quad-Level 1	RIMODE = GND	Fixed Off-Time	Automatic Retry	Overcurrent Only
Quad-Level 2	RIMODE = 20 kΩ to GND	Cycle-By-Cycle	Automatic Retry	Current Chopping and Overcurrent
Quad-Level 3	RIMODE = 62 kΩ to GND	Cycle-By-Cycle	Outputs Latched Off	Current Chopping and Overcurrent
Quad-Level 4	RIMODE = HI-Z	Fixed Off-Time	Outputs Latched Off	Overcurrent Only

*IMode can only be selected with the DRV8874. IMode can not be selected with the DRV8871; this IC supports Quad-Level 1 only
 *Pololu and AliExpress DRV8874 Modules include 20K SMD resistor on board (Quad-Level 2)



Alternative 1: DRV8874

Alternative 2: DRV8871

DRV8871 modules have a different (larger) footprint, and do not support a nFault output.

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Sheet: /CDE-Interface/
 File: CDE-Interface.kicad_sch

Title: LZ210-TMC

Size: A4 Date: 2026-09-16

Rev: V2.2

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Id: 7/10

V2.2: 3V3 zener removed.
 R704/R705 values decreased, C701 increased

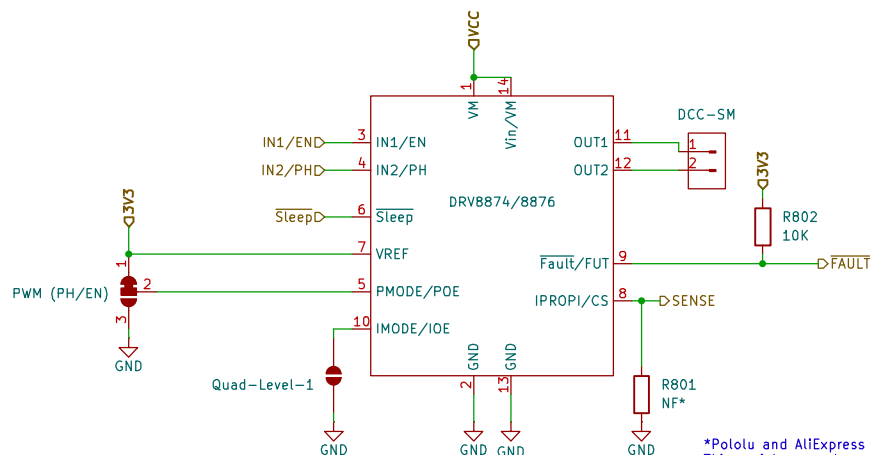
PMODE STATE	CONTROL MODE
PMODE = Logic Low	PH/EN
PMODE = Logic High	PWM
PMODE = Hi-Z	Independent Halfbridge

*DRV8871: only PWM

IMODE STATE		IMODE FUNCTION		nFAULT Response
Quad-Level	RIMODE	Current Chopping Mode	Overcurrent Response	
Quad-Level 1	RIMODE = GND	Fixed Off-Time	Automatic Retry	Overcurrent Only
Quad-Level 2	RIMODE = 20 kΩ to GND	Cycle-By-Cycle	Automatic Retry	Current Chopping and Overcurrent
Quad-Level 3	RIMODE = 62 kΩ to GND	Cycle-By-Cycle	Outputs Latched Off	Current Chopping and Overcurrent
Quad-Level 4	RIMODE = Hi-Z	Fixed Off-Time	Outputs Latched Off	Overcurrent Only

*Quad-Level 2 is chosen to facilitate inrush protection (3μs pulses)

*Pololu and AliExpress Modules include 20K SMD resistor on board (Quad-Level 2)



*Pololu and AliExpress Modules include a 2.49 KΩ on board SMD resistor between IPROPI and GND. This resistor may be replaced to obtain other Itrip / Vsense values. See table below.

Formula: $I_{TRIP} (A) \times AIPROPI (\mu A/A) = V_{REF} (V) / R_{IPROPI} (\Omega)$
 DRV8874: AIPROPI = 450 μA/A
 DRV8876: AIPROPI = 1000 μA/A

Chip (Vref = 3V3)	Rsense	Itrip	Vsense – 60mA ACK	Remarks
DRV8874	2.49 kΩ	2945 mA	67 mV	Fitted on module
DRV8874	6.8 kΩ	1078 mA	184 mV	
DRV8874	15 kΩ	489 mA	405 mV	
DRV8874	27 kΩ	272 mA	729 mV	
DRV8876	2.49 kΩ	1325 mA	149 mV	Fitted on module
DRV8876	6.8 kΩ	485 mA	408mV	
DRV8876	15 kΩ	220 mA	900 mV	

According to RCN216, the max. current should be limited to 250 ma (+/- 20%)

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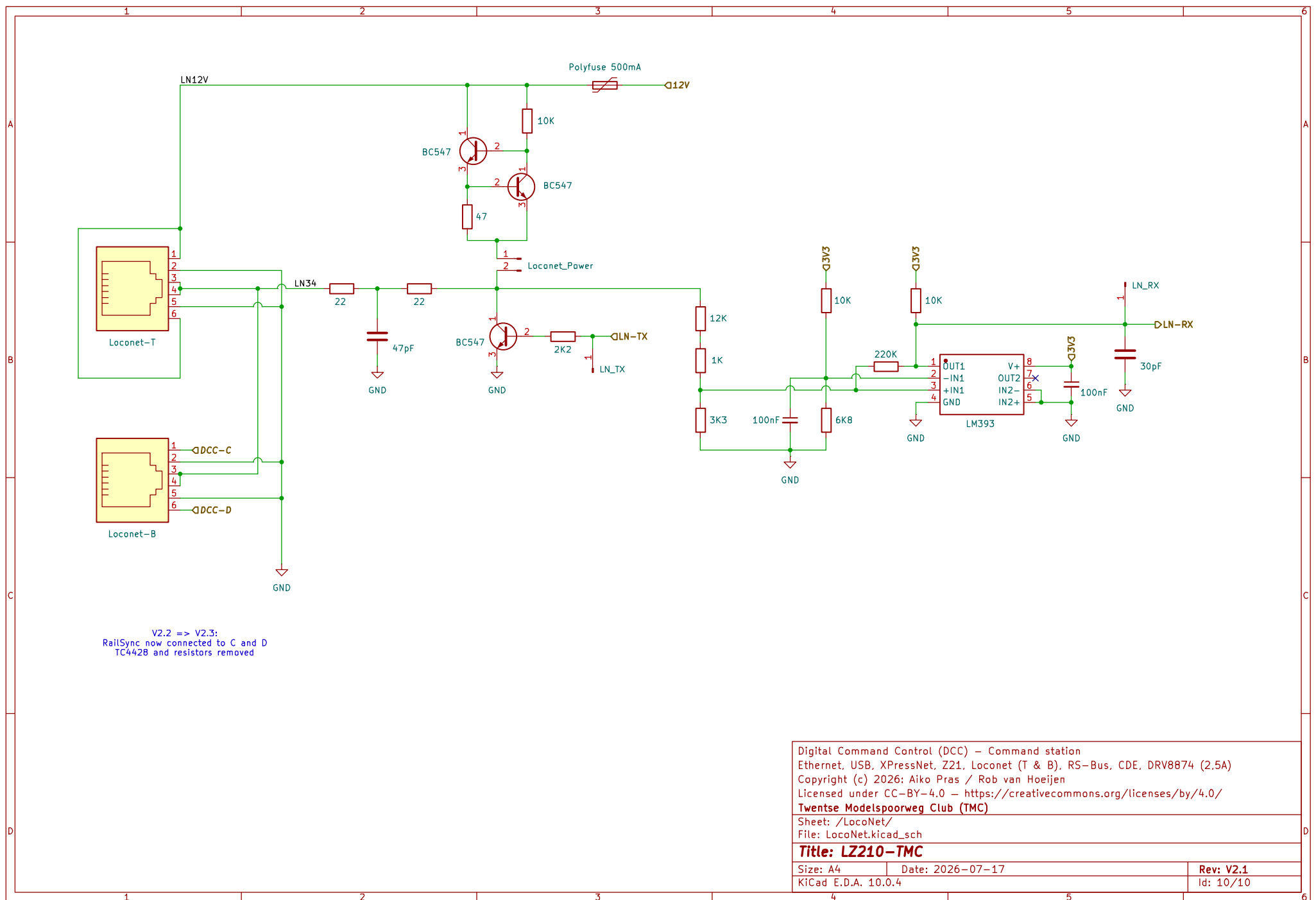
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Sheet: /ProgrammingTrack/
 File: ProgrammingTrack.kicad_sch

Title: LZ210-TMC

Size: A4	Date: 2026-07-17	Rev: V2.1
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Sheet: /LocoNet/
 File: LocoNet.kicad_sch

Title: LZ210-TMC

Size: A4 Date: 2026-07-17

Rev: V2.1

KiCad E.D.A. 10.0.4

Id: 10/10